「Def.

A group action on G acts G as

$$G \times G \to G : (g, a) \mapsto g \cdot a$$

is called acts by Conjugation.

And, the orbits of G acting on itself by Conjugation are called <u>Conjugacy classes.</u>

「Def.

A group action on G acts on the Power set $\mathcal{P}(G)$ as

$$G \times \mathcal{P}(G) : (g, A) \mapsto gAg^{-1}$$

is called acts by Conjugate on the power set.

「Prop. The number of Conjugates of a subset $S \subseteq G$ is $|G : N_G(S)|$.

Proof. Generally, for the stabilizer G_a and orbit C_a of a, $|C_a| = |G : G_a|$, because

$$C_a \to \{gG_a \mid g \in G\} : g \cdot a \mapsto gG_a \left(= g\{g' \in G \mid g' a = a\}\right)$$

is well-defined and injective by

$$g_1 \cdot a = g_2 \cdot a \iff g_1^{-1} g_2 \cdot a = a \iff g_1^{-1} g_2 \in G_a \iff g_1 G_a = g_2 G_a$$

And surjective because given $g \in G$, $g \cdot a \in C_a$. Thus, since

$$G_S = \{g \in G \mid gSg^{-1} = S\} = N_G(S)$$

gives the result.